

```
rbhanu_priya@LAPTOP-7PGTCMPB:~$ cd OSLab
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab$ cd fork
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ nano pro.c
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ gcc pro.c -o pro
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./pro.c
-bash: ./pro.c: Permission denied
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./pro
Enter a Linux command: uname
```

```
Parent Process
Parent PID : 645
Child PID  : 646
```

```
Child Process
Child PID : 646
Parent PID: 645
Linux
```

```
Child process completed.
```

```
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ |
```

```
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ^C
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./pro
Enter a Linux command: lscpu
```

```
Parent Process
Parent PID : 666
Child PID  : 669
```

```
Child Process
Child PID : 669
Parent PID: 666
```

```
Architecture:          x86_64
  CPU op-mode(s):      32-bit, 64-bit
  Address sizes:        39 bits physical, 48 bits virtual
  Byte Order:           Little Endian
CPU(s):                12
  On-line CPU(s) list: 0-11
Vendor ID:             GenuineIntel
  Model name:           13th Gen Intel(R) Core(TM) i5-1334U
    CPU family:         6
    Model:              186
  Thread(s) per core:   2
  Core(s) per socket:   6
  Socket(s):            1
  Stepping:             3
  BogomIPS:             4992.00
  Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                        r pge mca cmov pat pse36 clflush mmx fxsr sse s
                        se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                        _tsc rep_good nopl xtopology tsc_reliable nonst
                        op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                        sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                        opcnt tsc_deadline_timer aes xsave avx f16c rdr
                        and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                        brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                        d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                        2 erms invpcid rdseed adx smap clflushopt clwb
                        sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                        vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                        vdiri movdir64b fsrm md_clear serialize ibt flu
                        sh_lld arch_capabilities
```

```
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:              Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DISABLE
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
  Vmscape:                Not affected

Child process completed.
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ |
```

```
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./pro
```

```
Enter a Linux command: lsblk
```

```
Parent Process
```

```
Parent PID : 679
```

```
Child PID  : 721
```

```
Child Process
```

```
Child PID : 721
```

```
Parent PID: 679
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	356.9M	1	disk	
sdb	8:16	0	159.4M	1	disk	
sdc	8:32	0	2G	0	disk	[SWAP]
sdd	8:48	0	1T	0	disk	/mnt/wslg/distro
						/

```
Child process completed.
```

```
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ |
```

```
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ ./pro
```

```
Enter a Linux command: ps
```

```
Parent Process
```

```
Parent PID : 731
```

```
Child PID  : 732
```

```
Child Process
```

```
Child PID : 732
```

```
Parent PID: 731
```

PID	TTY	TIME	CMD
563	pts/0	00:00:00	bash
731	pts/0	00:00:00	pro
732	pts/0	00:00:00	ps

```
Child process completed.
```

```
rbhanu_priya@LAPTOP-7PGTCMPB:~/OSLab/fork$ |
```

Child Process

Parent Process

Parent PID : 742

Child PID : 743

Child PID : 743

Parent PID: 742

top - 06:52:06 up 5 min, 1 user, load average: 0.02, 0.01, 0.00

Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.

MiB Mem : 7762.6 total, 6336.7 free, 545.2 used, 1033.1 buff/cache

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7217.5 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	24092	15368	11584	S	0.0	0.2	0:00.54
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01
8	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00
48	root	19	-1	50432	16748	15484	S	0.0	0.2	0:00.15
80	systemd+	20	0	22416	14480	11984	S	0.0	0.2	0:00.07
87	root	20	0	35380	12348	9240	S	0.0	0.2	0:00.33
145	root	20	0	2888	1944	1820	S	0.0	0.0	0:00.03
146	root	20	0	4472	3156	2912	S	0.0	0.0	0:00.01
147	message+	20	0	8820	5368	4692	S	0.0	0.1	0:00.04
155	root	20	0	44096	29848	14712	S	0.0	0.4	0:00.21
165	root	20	0	18436	9412	8168	S	0.0	0.1	0:00.07
189	root	20	0	1866288	15364	12552	S	0.0	0.2	0:00.13
212	syslog	20	0	220548	5956	4656	S	0.0	0.1	0:00.09
265	_chrony	20	0	23692	11236	7240	S	0.0	0.1	0:00.05
268	_chrony	20	0	12096	2428	1784	S	0.0	0.0	0:00.00
276	root	20	0	123460	32732	18116	S	0.0	0.4	0:00.13
279	root	20	0	5492	2736	2556	S	0.0	0.0	0:00.00
328	root	20	0	8652	5612	4764	S	0.0	0.1	0:00.01
370	rbhanu_+	20	0	22760	13452	10992	S	0.0	0.2	0:00.14
372	rbhanu_+	20	0	22916	4092	2088	S	0.0	0.1	0:00.00
398	rbhanu_+	20	0	6136	5416	3848	S	0.0	0.1	0:00.02
561	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00
562	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.03
563	rbhanu_+	20	0	6268	5540	3864	S	0.0	0.1	0:00.09
742	rbhanu_+	20	0	2768	1872	1776	S	0.0	0.0	0:00.00